

First Problem Sheet

on
Measure Theory

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Topics: Set Theory
Maps
Countability

“The essence of mathematics
lies in its freedom.”
-Georg Cantor

Problem 1

If A and B are sets, show that $A \subseteq B$ if and only if $A \cap B = A$.

Problem 2

If A is a set and $\mathcal{B} = \{B_i : i \in I\}$ is any family of sets, show that

$$A \cap \bigcup_{i \in I} B_i = \bigcup_{i \in I} (A \cap B_i) \quad \text{and} \quad A \cup \bigcap_{i \in I} B_i = \bigcap_{i \in I} (A \cup B_i).$$

Problem 3

Let B be any set. Show that $(A \cap B) \cup C = A \cap (B \cup C)$ if and only if $C \subseteq A$.

Problem 4

The **symmetric difference** of two sets A and B is the set D of all elements that belong to either A or B but not both, symbolically, $D \equiv A \Delta B = (A \cup B) \setminus (A \cap B)$. Represent D with a diagram and show that $D = (A \setminus B) \cup (B \setminus A)$. Furthermore, show that

- i) $A \subseteq A \Delta B \cup A$,
- ii) $A \subseteq A \Delta B \cup B$,
- iii) $B \subseteq A \Delta B \cup A$,
- iv) $B \subseteq A \Delta B \cup B$.

Problem 5

Show that if $f : A \rightarrow B$ and E, F are subsets of A , then

$$f(E \cup F) = f(E) \cup f(F) \quad \text{and} \quad f(E \cap F) \subseteq f(E) \cap f(F).$$

Problem 6

Show that if $f : A \rightarrow B$ and G, H are subset of B , then $f^{-1}(G \cup H) = f^{-1}(G) \cup f^{-1}(H)$ and $f^{-1}(G \cap H) = f^{-1}(G) \cap f^{-1}(H)$

Problem 7

Show that if $f : A \rightarrow B$ is injective and $E \subseteq A$ [respectively, surjective and $H \subseteq B$], then $f^{-1}(f(E)) = E$ [respectively, $f(f^{-1}(H)) = H$].

Problem 8

Suppose that f is an injection. Show that $(f^{-1} \circ f)(x) = x$ for all $x \in \text{Dom}(f)$ and that $(f \circ f^{-1})(y) = y$ for all $y \in R(f)$.

Problem 9

Show that if f is a bijection from A onto B , then f^{-1} is a bijection of B onto A . Moreover, if g is a bijection of B onto C , then the composite function $g \circ f$ is a bijection of A onto C .

Problem 10

Prove Theorem 1.3.

Problem 11

A complex number $z \in \mathbb{C}$ is said to be an **algebraic number** if it is the root of some nonzero polynomial with integer coefficients. Show that the set of all algebraic numbers is denumerable.